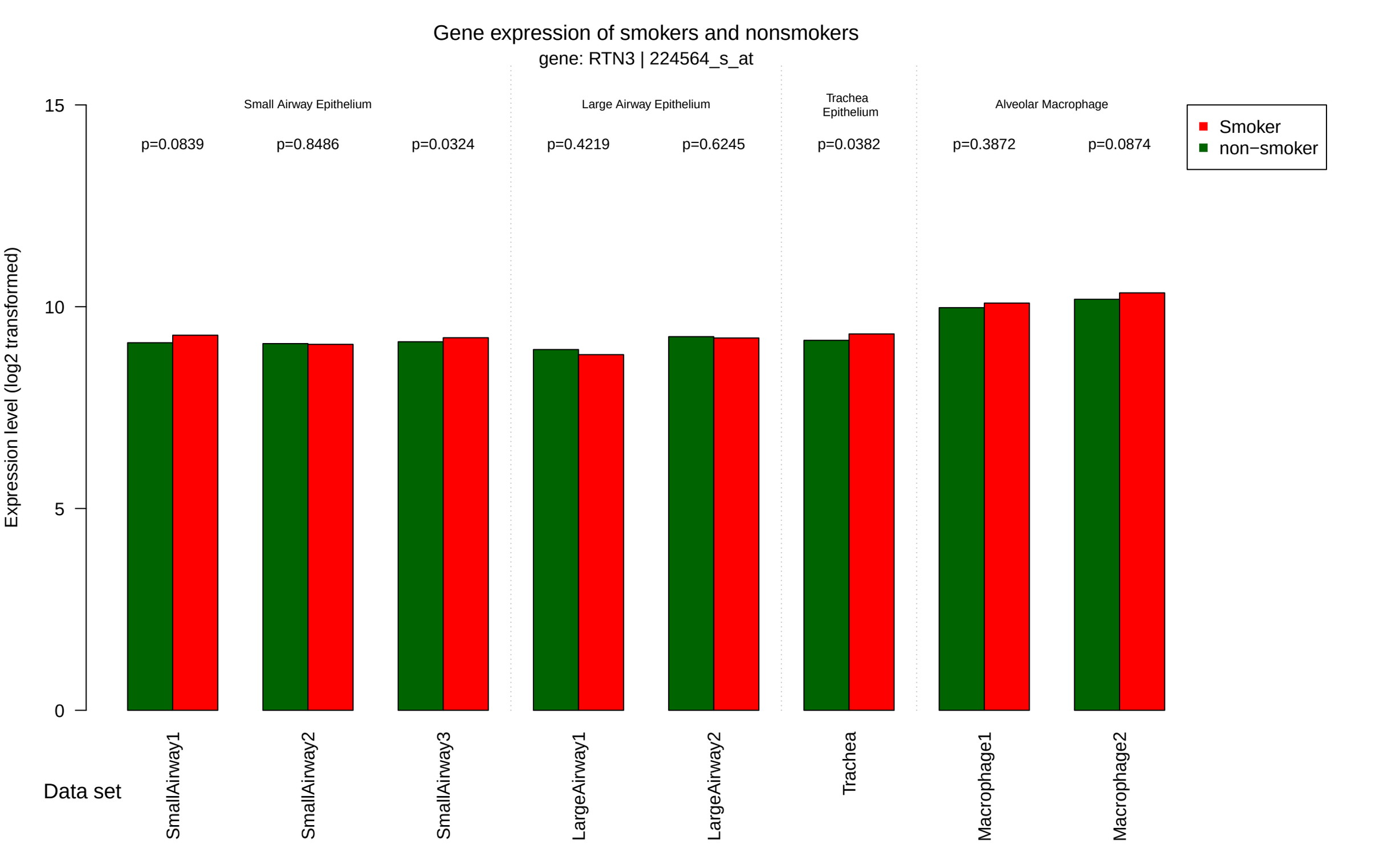
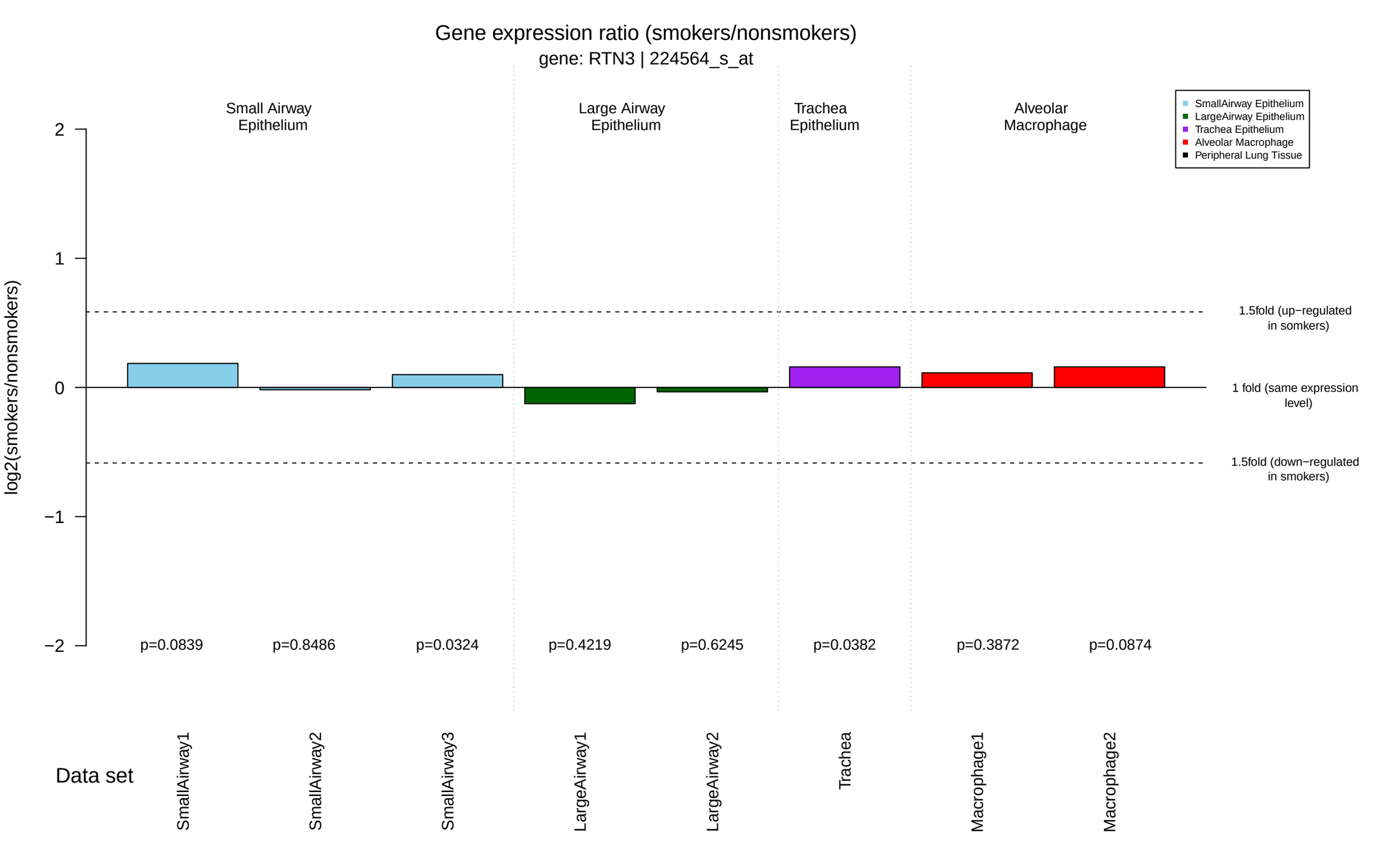
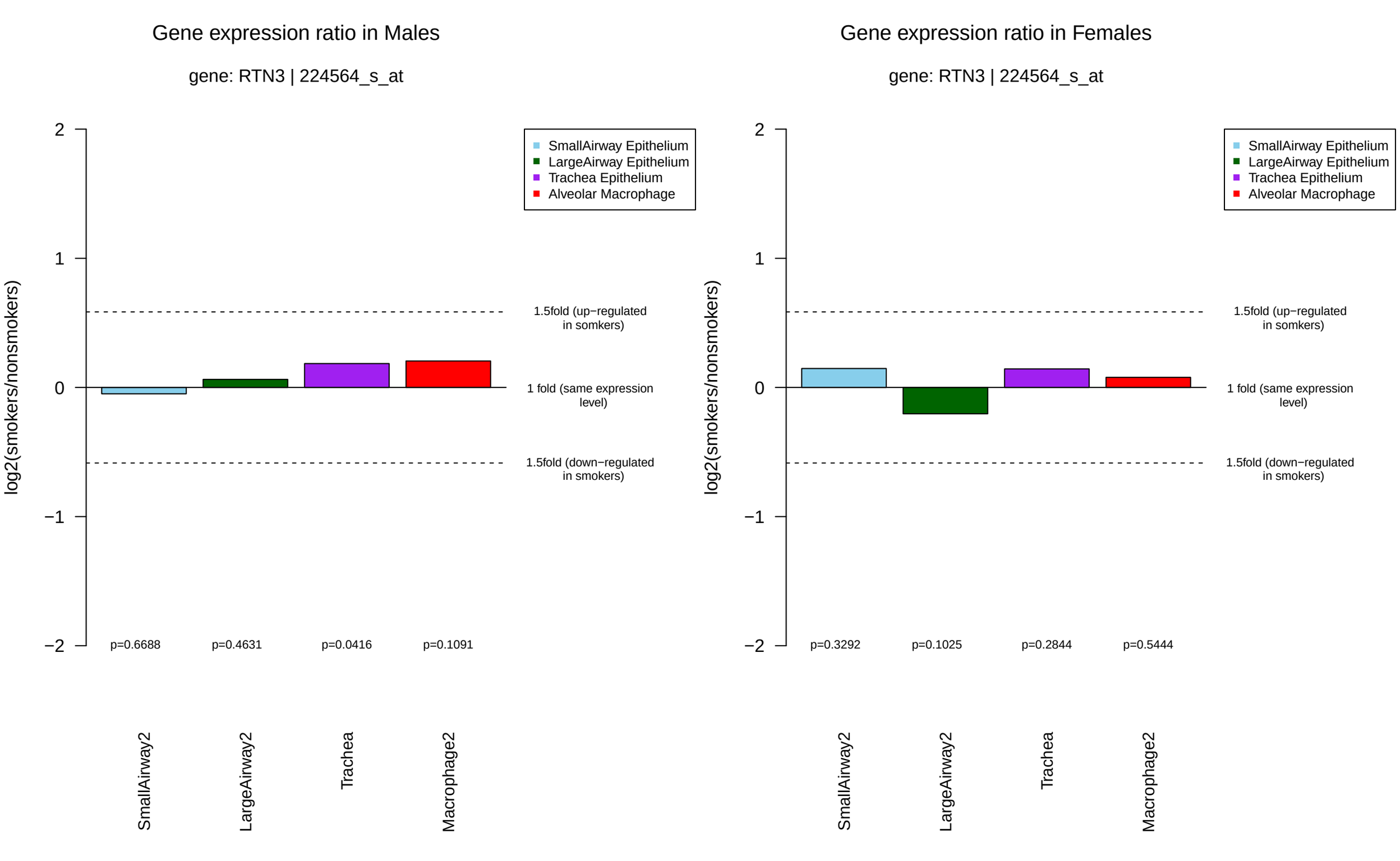
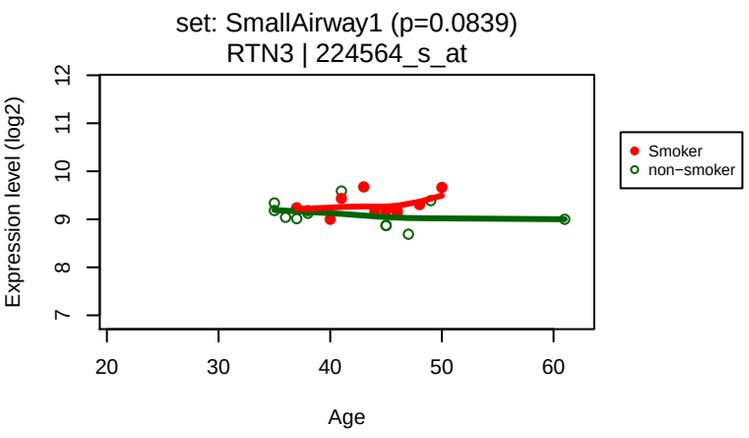
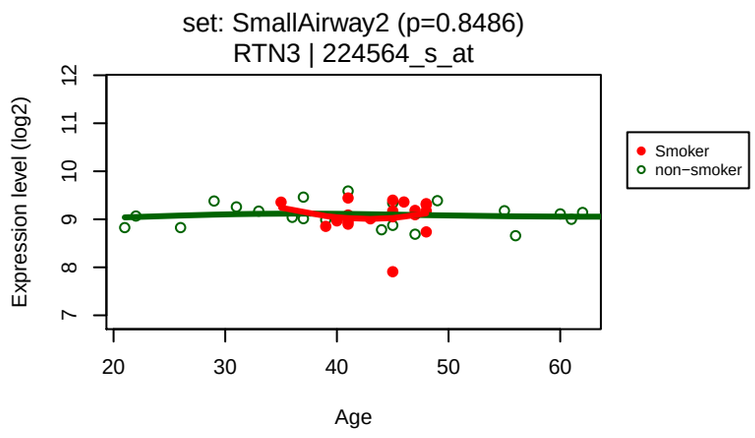
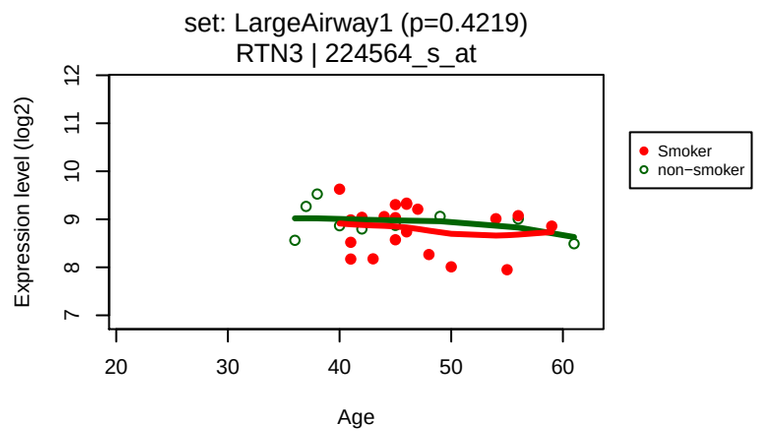

Gene expression of smokers and nonsmokers
gene: RTN3 | 224564_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0839
p=0.8486
p=0.0324
p=0.4219
p=0.6245
p=0.0382
p=0.3872
p=0.0874
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

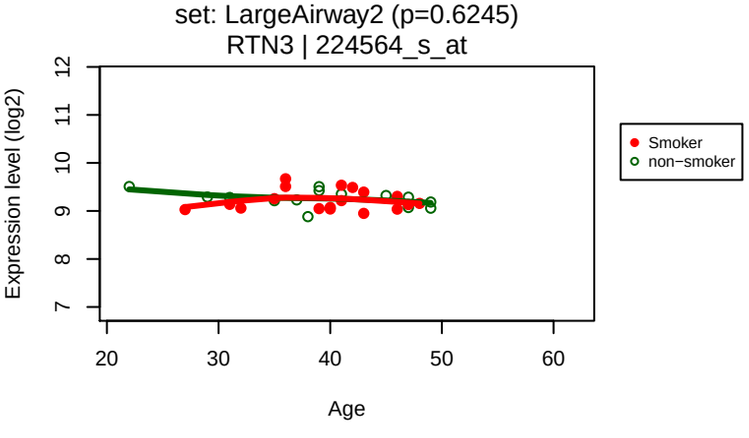

Gene expression ratio (smokers/nonsmokers)
gene: RTN3 | 224564_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0839
p=0.8486
p=0.0324
p=0.4219
p=0.6245
p=0.0382
p=0.3872
p=0.0874
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

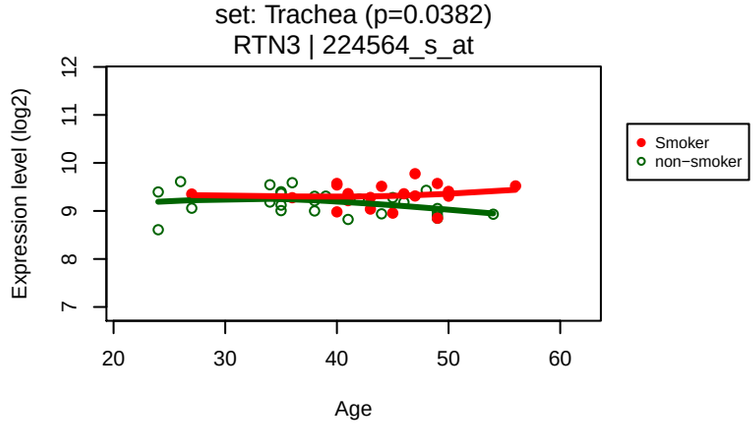

Gene expression ratio in Males
gene: RTN3 | 224564_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6688
p=0.4631
p=0.0416
p=0.1091
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RTN3 | 224564_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3292
p=0.1025
p=0.2844
p=0.5444
SmallAirway2
LargeAirway2
Trachea
Macrophage2

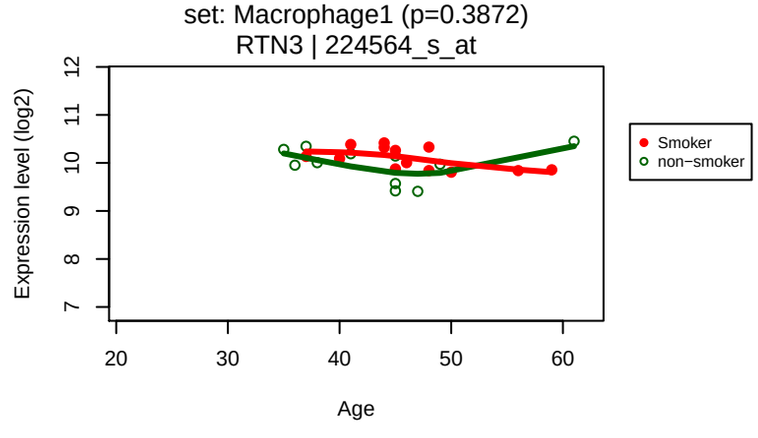

set: SmallAirway1 (p=0.0839)
RTN3 | 224564_s_at
Expression level (log2)
Age
Smoker
non−smoker

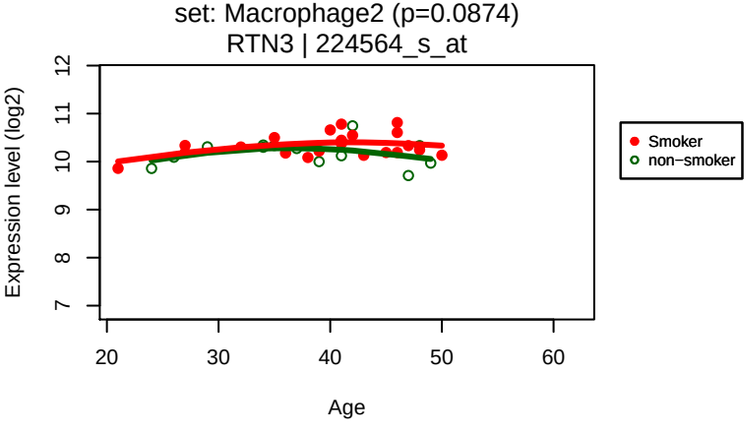

set: SmallAirway2 (p=0.8486)
RTN3 | 224564_s_at
Expression level (log2)
Age
Smoker
non−smoker

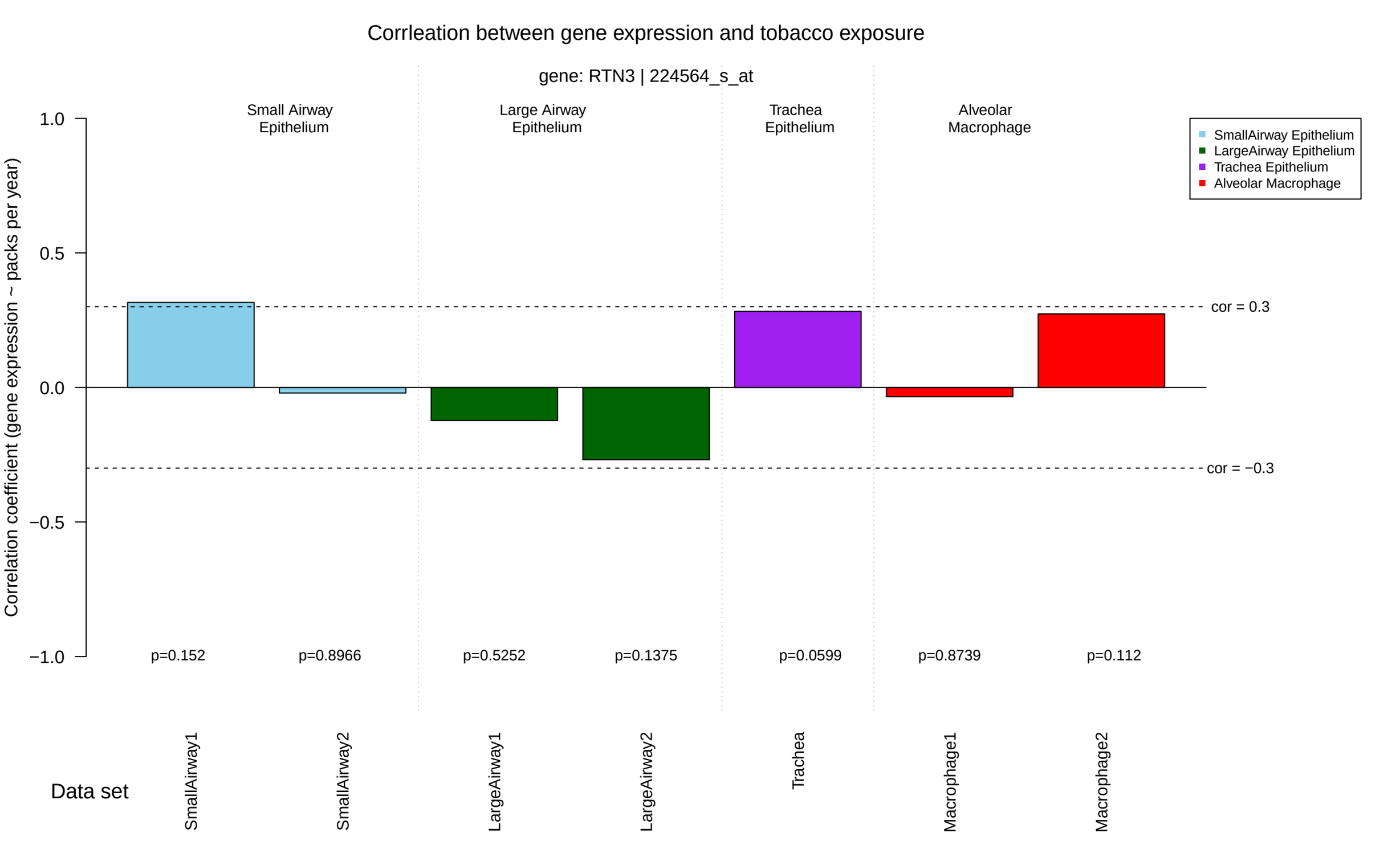

set: LargeAirway1 (p=0.4219)
RTN3 | 224564_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6245)
RTN3 | 224564_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0382)
RTN3 | 224564_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3872)
RTN3 | 224564_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0874)
RTN3 | 224564_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RTN3 | 224564_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.152
p=0.8966
p=0.5252
p=0.1375
p=0.0599
p=0.8739
p=0.112
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2